Spatiotemporal forecast with deep learning

Disciplina: Programação aplicada à engenharia cartográfica

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26 de outubro de 2026

Sumário



1. Introduction
2. Data set construction
3. Architectures
4. State of the art (2026)
5. Hands on
6. Other topics
7. Referências

Sumário

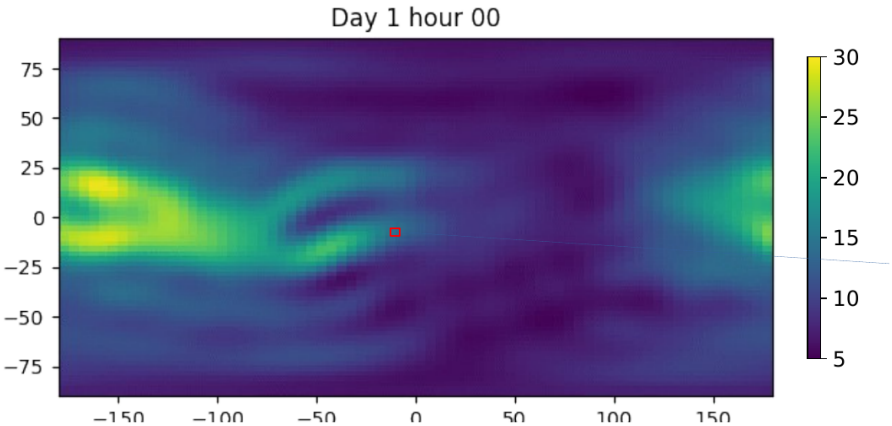


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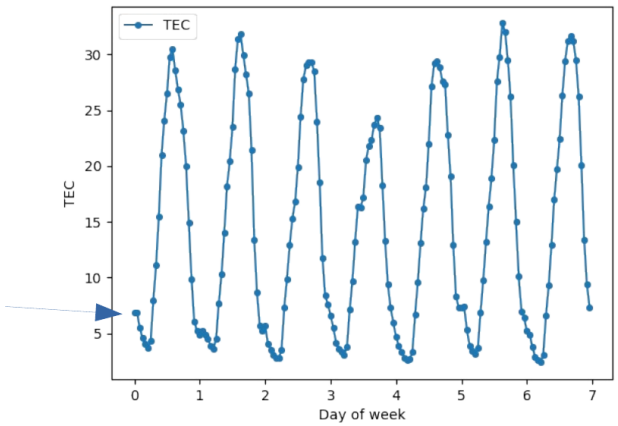
Global ionosphere maps prediction

Objective: Share my experience with deep learning for ionosphere maps forecast.

Predict the next maps in a time series.



Ionosphere maps (VTEC) of a single day (single band, but colored).



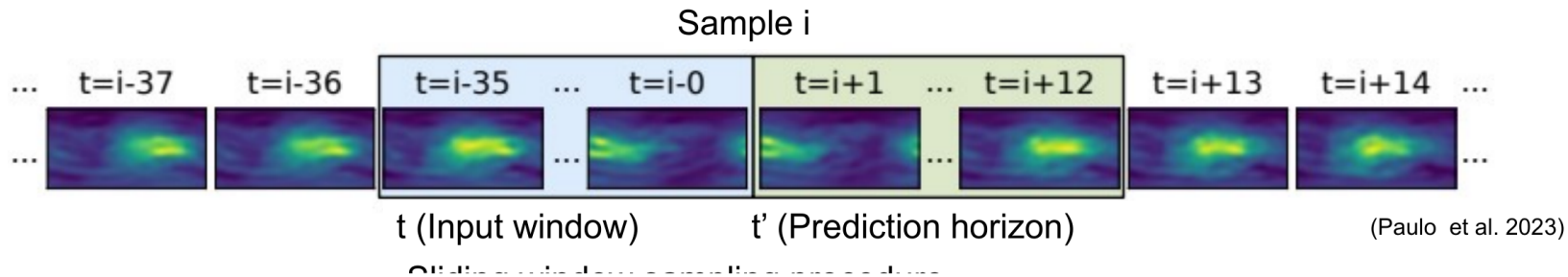
VTEC time series of a single pixel of the map along an entire week.

`https://github.com/mauriciodev/tec_forecast` (Tensorflow 2/Keras)

Paulo MCM de, Marques HA, Feitosa RQ, Ferreira MP (2023) New encoder–decoder convolutional LSTM neural network architectures for next-day global ionosphere maps forecast. GPS Solut 27(2):95.

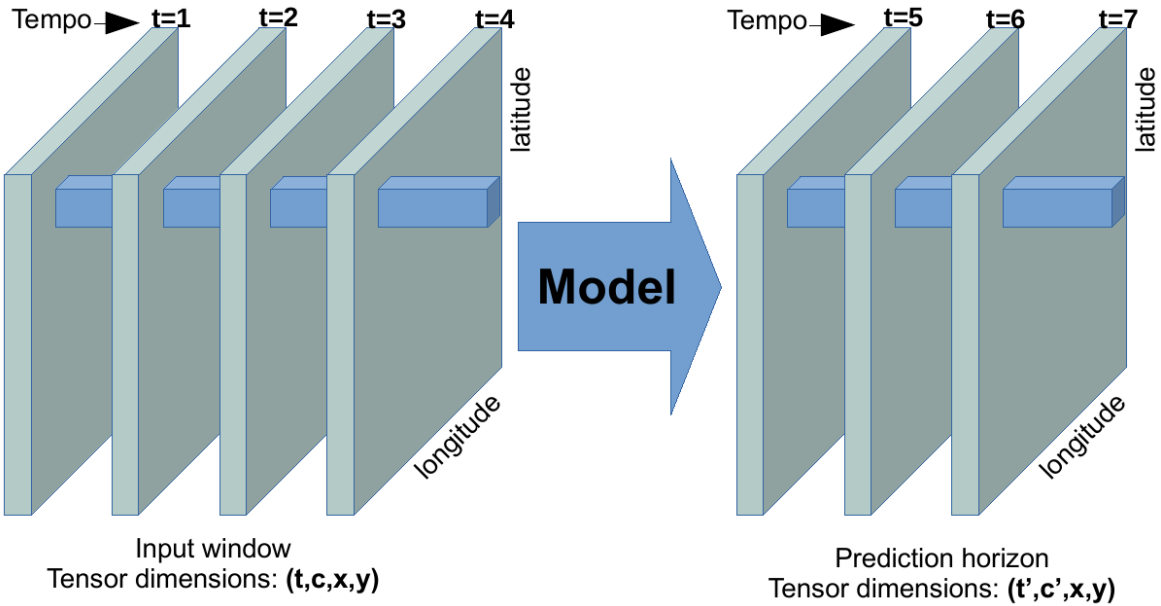
`https://doi.org/10.1007/s10291-023-01442-4`

Prediction of the next maps in a time series



(Paulo et al. 2023)

Sliding window sampling procedure.



Main loss functions:

$$MAE = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

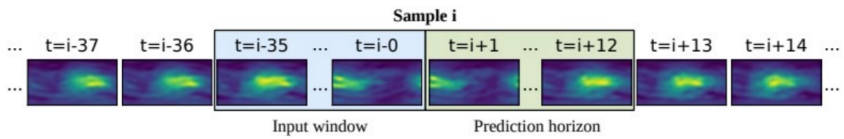
The model transforms from t input images to the next t' predicted images.

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Hyperparameters

- Sequences of images
  - Time, channels, x, y
- PyTorch
  - Input: (t, c, x, y)
  - Output: (t', c', x, y)



How do we slice it to produce samples?

(Paulo et al. 2023)

- **Input sequence length (t)**
  - Must be long enough to provide data for the prediction process.
- **Output sequence length (t')**
  - Must be long enough to be usable.
- **Spatial resolution**
  - Spatial tensor dimensions (x, y)
  - Multiples of 2^n, where n is the number of pooling layers.
- **Temporal resolution**
  - Must be compatible with the event.

Exploring the data

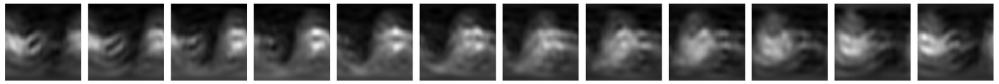
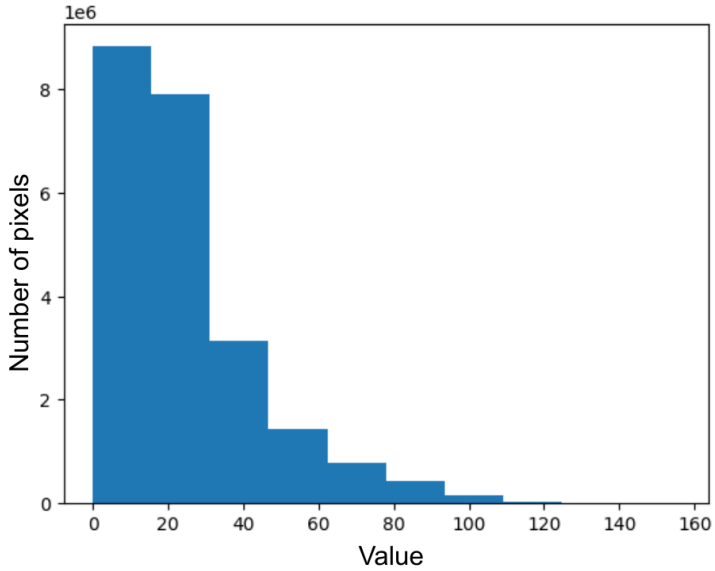


Image sequence.

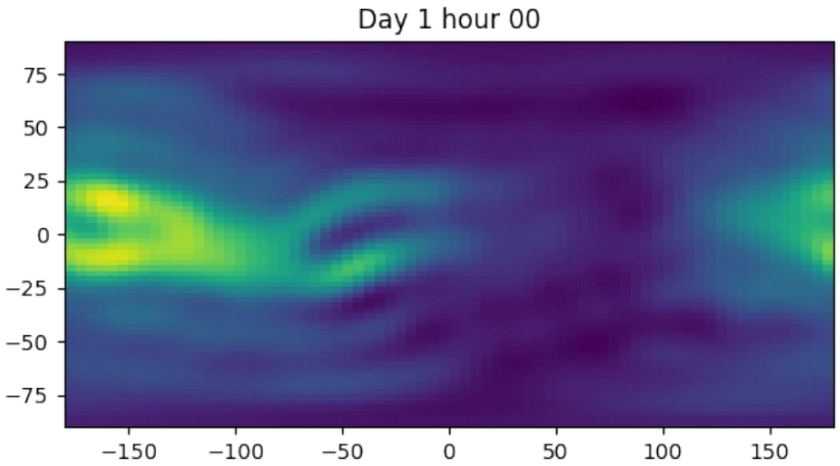
- Hyperparameters definition
  - Previous research
  - Trial and error
  - Hyperparameters search (Don't have to be fancy)
- Time dimension
  - Trend
  - Season
  - Noise
- Spatial dimensions
  - Spatial relations
  - Clusters, spatial trends, etc



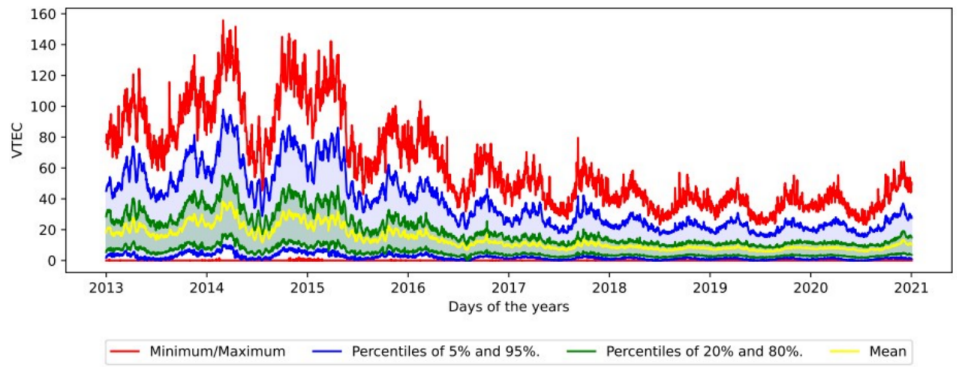
Flattened data histogram.

$(t, c, x, y) \rightarrow (t \cdot x \cdot y, c)$

Exploring the data



Video sequences can help understand the spatiotemporal trends.



Percentiles of the reshaped tensor.

$$(t, c, x, y) \rightarrow (t, c \cdot x \cdot y)$$

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Architectures

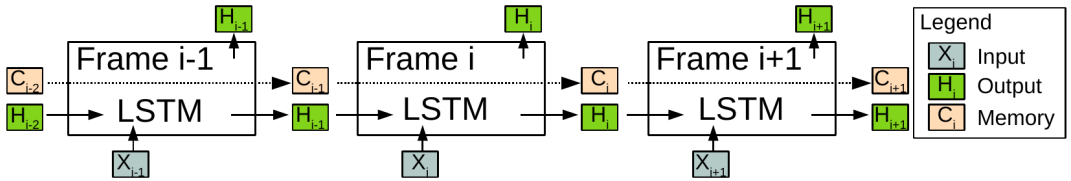


- Recurrent neural networks
- 3D Convolutional networks
- 2D Convolutional networks (based on SimVP)
- Transformers networks
- Structured state space sequence models (SSM)

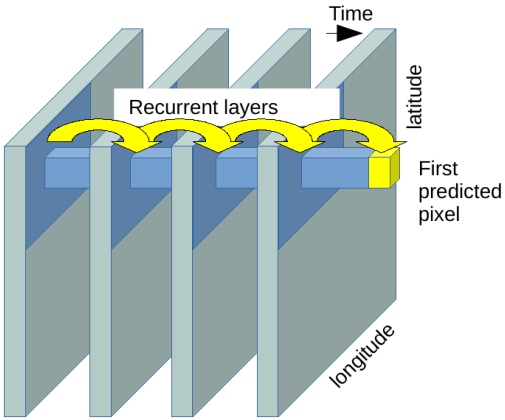
Recurrent neural networks: ConvLSTM

$H_i = f(X_i, H_{i-1}, C_{i-1})$
With: $H_0 = 0$ e $C_0 = 0$

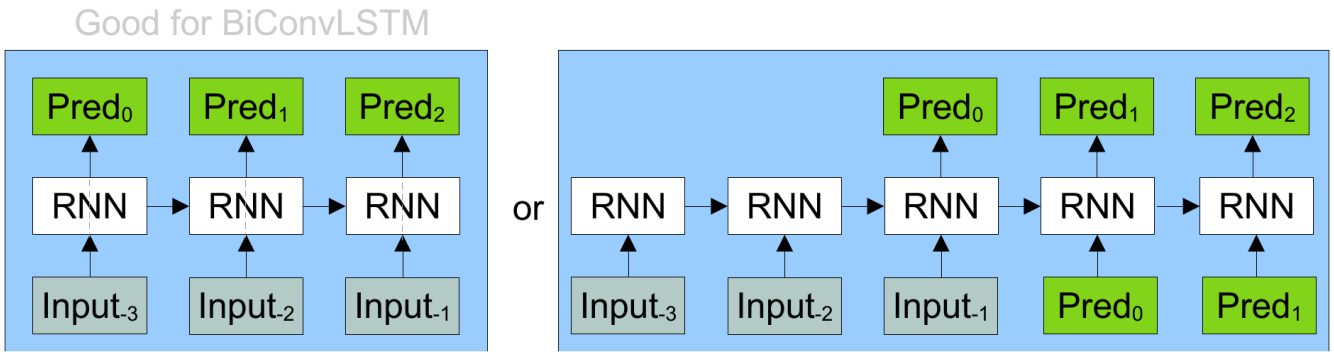
X_i: input data of current frame;
H_i: output of current frame;
H_{i-1}: output of the previous frame;
C_{i-1}: memory cell of the previous frame.



Recurrent LSTM cell processing for frames X_{i-1}, X_i e X_{i+1}



Single step prediction with RNN



Prediction methods with one RNN

Recurrent neural networks: ED-ConvLSTM

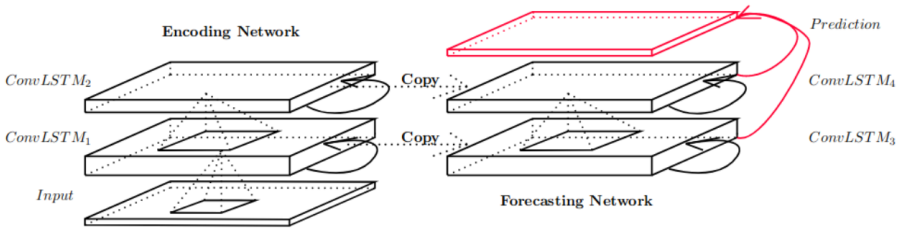
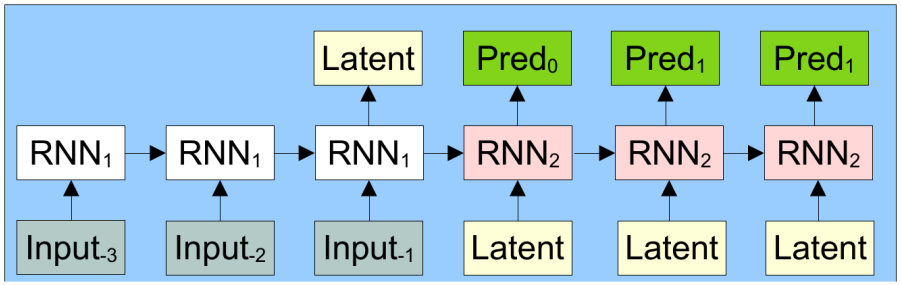


Figure 3: Encoding-forecasting ConvLSTM network for precipitation nowcasting

ConvLSTM was actually first designed with different layers for encoding and forecasting. (Shi et al. 2015)



Prediction method with Encoder-Decoder ConvLSTM.

Most articles ignore the difference between ConvLSTM and ED-ConvLSTM, but there are articles that show the error propagation.

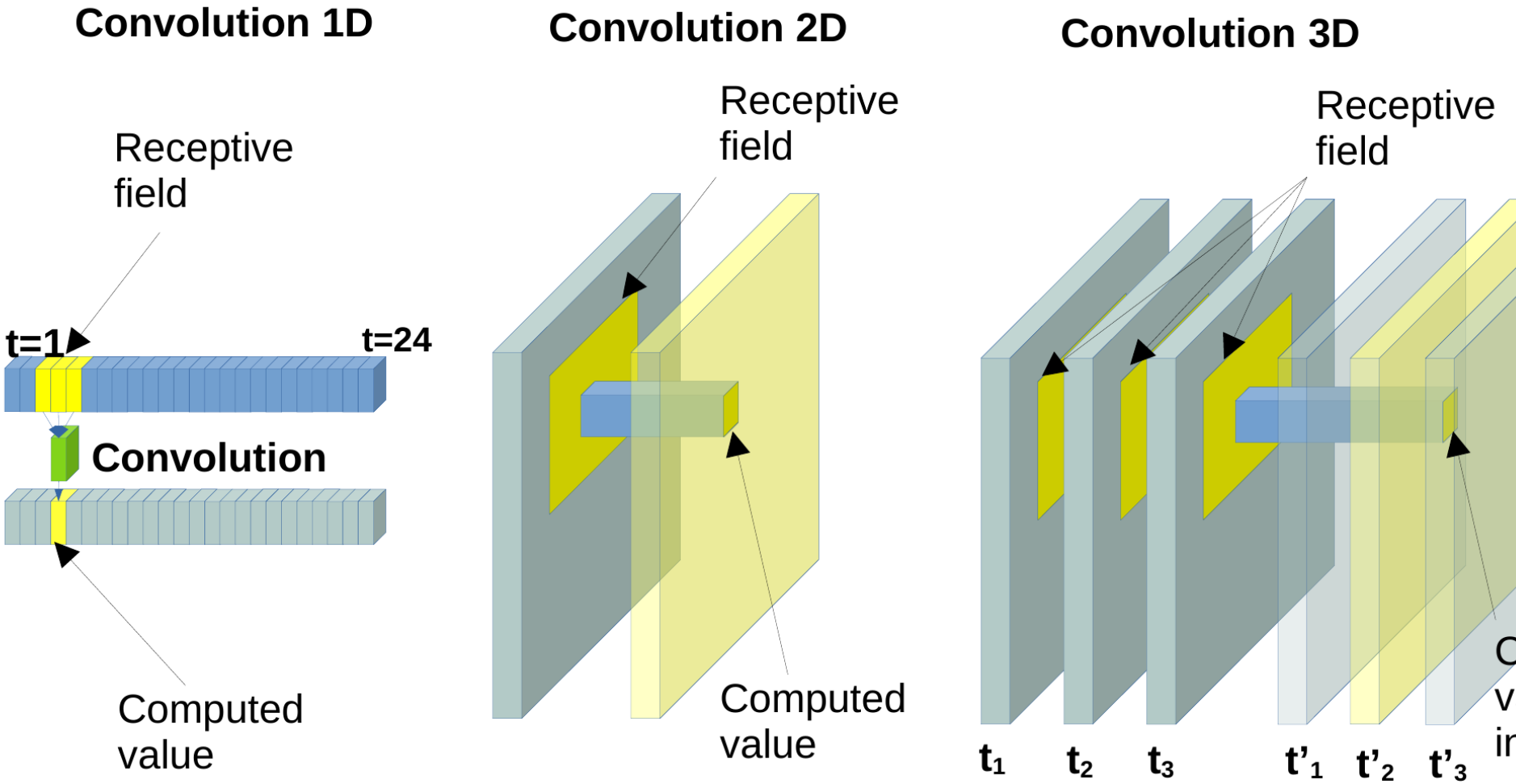
Recurrent Neural Networks evolution



- ConvLSTM (2015)
  - Uses a Memory Cell (C) that stores information through time steps.
  - 2D convolutions on each frame.
- PredRNN (2017) extends ConvLSTM
  - ST-LSTM: Adds a new memory tensor (M) that is passed through stacked RNN layers.
- E3D-LSTM (2018) extends PredRNN
  - 3D Convolutions instead of 2D.
  - Attention mechanism on tensor C.
- PredRNN v2 (2022) extends PredRNN
  - Adds a decoupled loss to learn different features with C and M.
- **Future?: XLSTM (2024) there is no ConvXLSTM yet.**

References can be found in OpenSTL GitHub page.

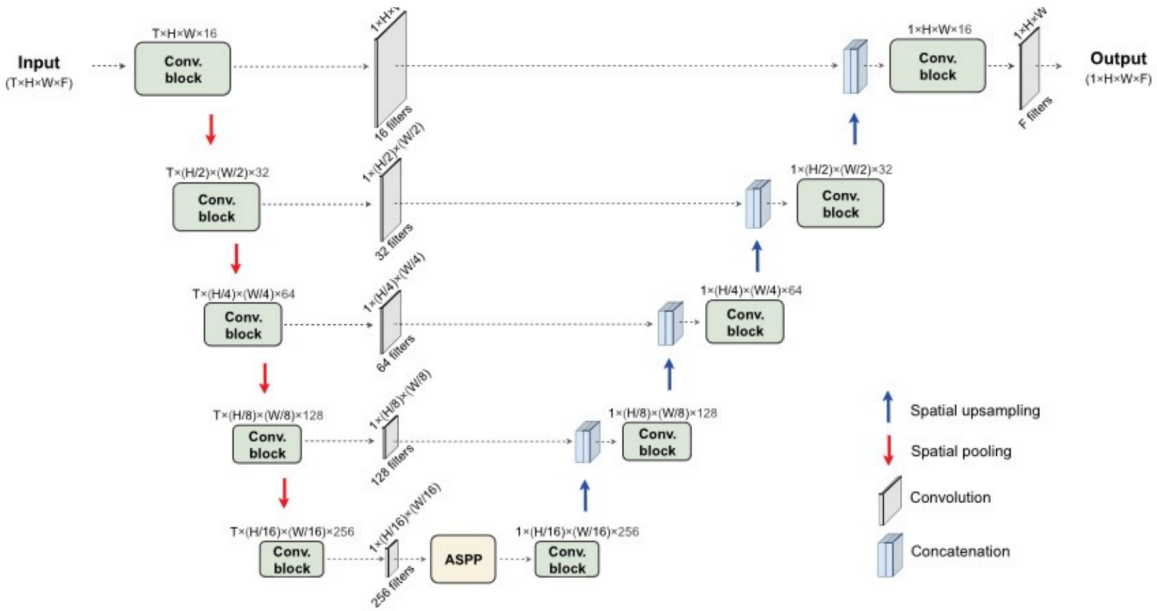
Remembering convolutional layers

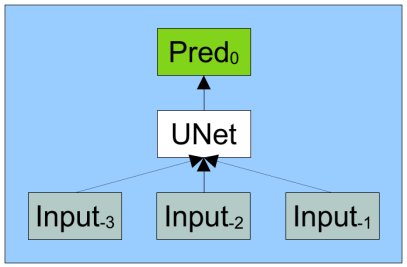


Obs.: 3D convolutions with stride 1 and pad do not modify the t, x, y dimensions.

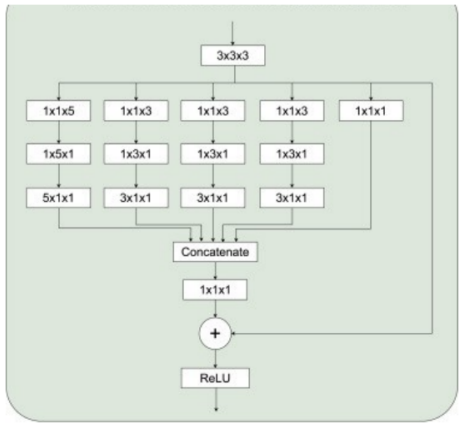
3D Convolutional networks: Broad-UNet (3D)

- Broad-UNet: Multi-scale feature learning for nowcasting tasks
- `https://github.com/jesusgf96/Broad-UNet`



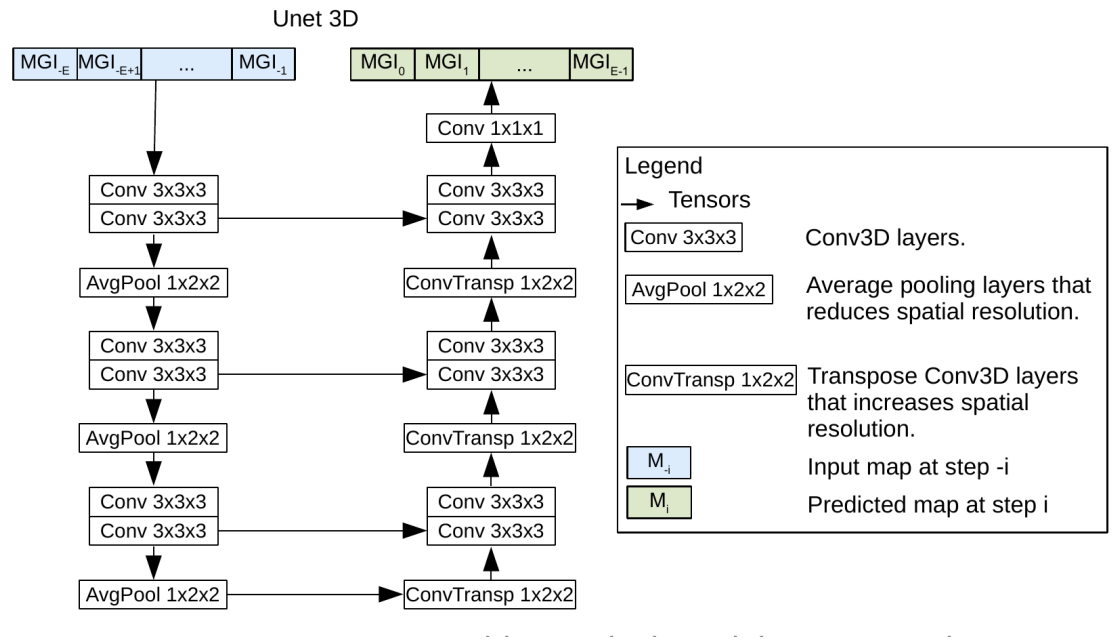


Prediction method with Broad-Unet 3D



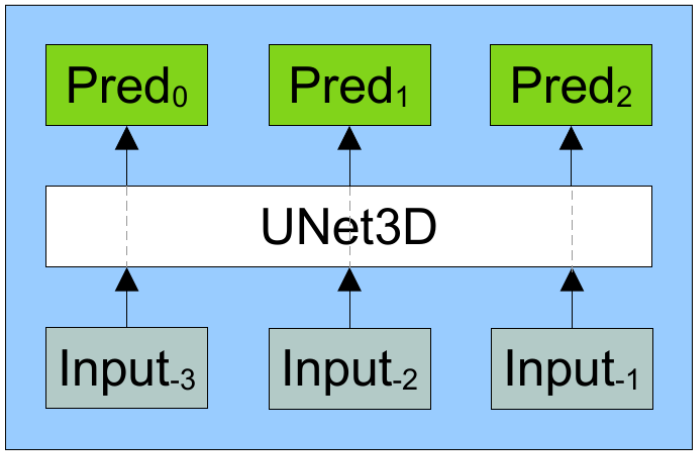
(Fernández and Mehrkanoon 2021)

3D Convolutional networks: UNet 3D



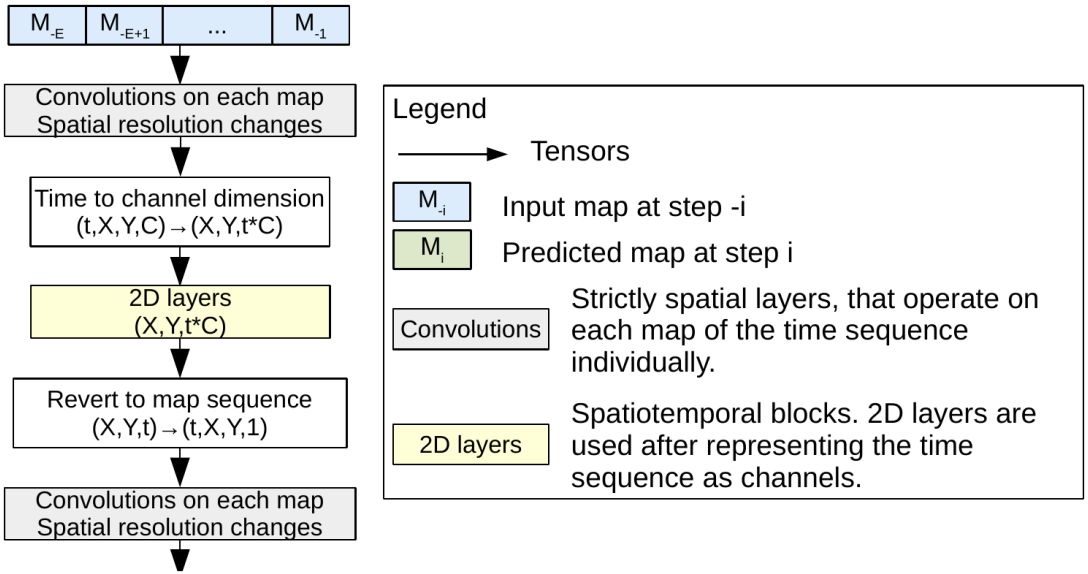
UNet3D architecture: hierarquical spatial representation widens the spatial receptive field of each predicted pixel.

Remember BiConvLSTM?

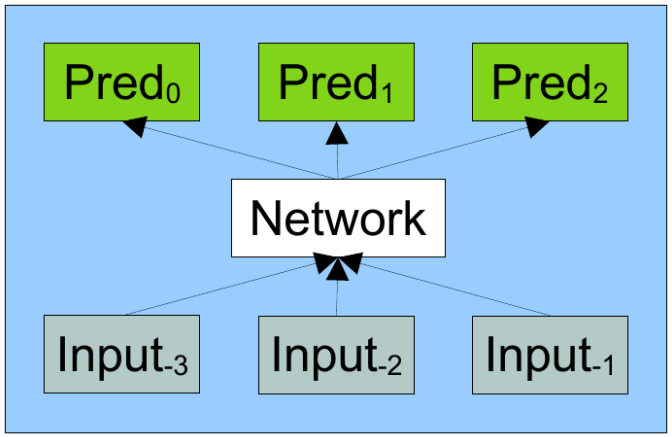


Prediction method with UNet3D

2D Convolutional networks: SimVP v2



SimVP architecture (TAN et al., 2022): The time dimension is represented in the channel dimension, thus 2D layers can be used to processes sequences of maps.

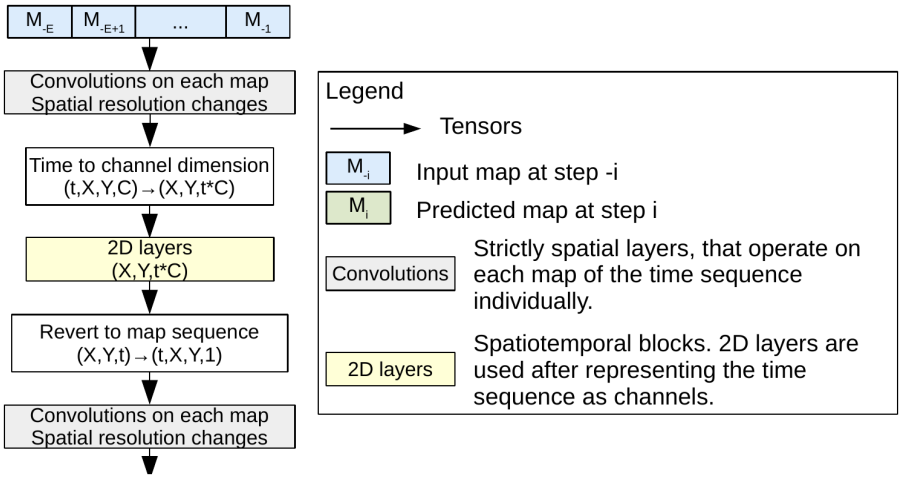


Prediction method with SimVP

The receptive field in the time dimension is the entire time sequence for each pixel.

Downside: Too many values on each neuron.

2D Convolutional networks: SimVP v2



SimVP architecture (TAN et al., 2022): The time dimension is represented in the channel dimension, thus 2D layers can be used to processes sequences of maps.

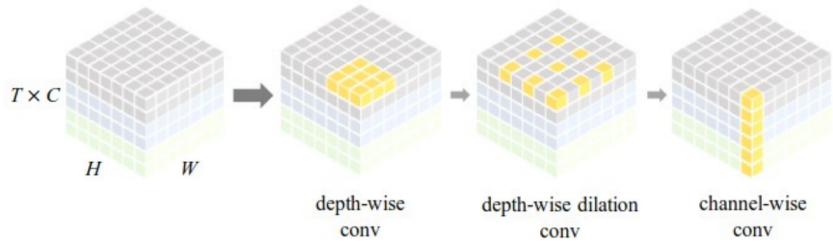
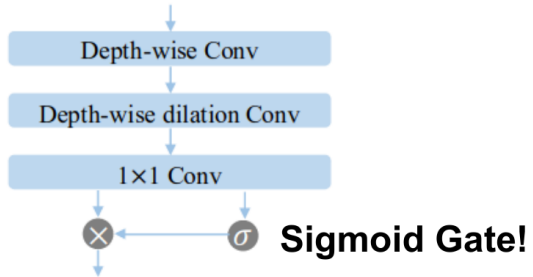


Fig. 7. The large kernel convolution in the gated spatiotemporal attention module. The yellow region denotes the receptive field.

Gated Spatiotemporal Attention Translator (gSTA)

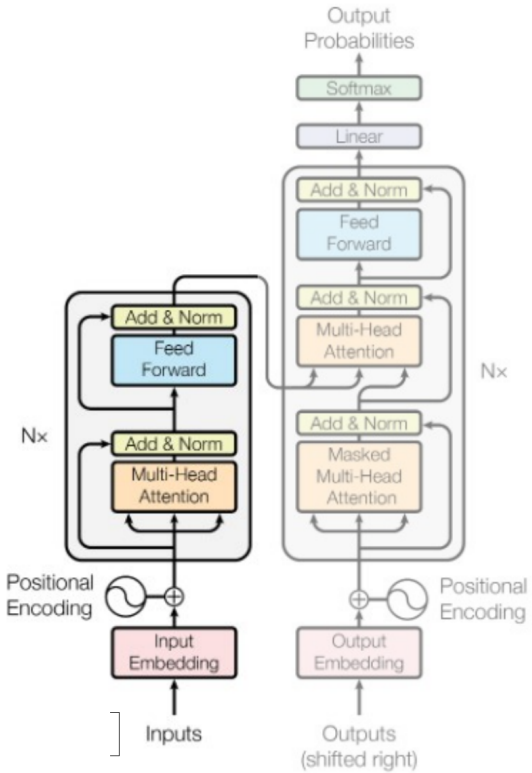
`https://github.com/chengtan9907/OpenSTL`



(Tan et al. 2022)

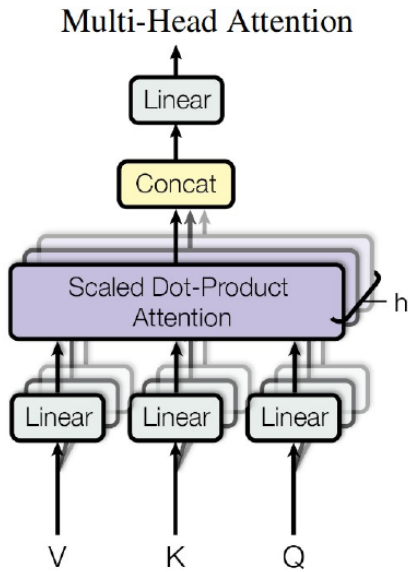
Remembering Transformer concepts

Natural Language processing
Memory: $O(\text{tokens}^2)$

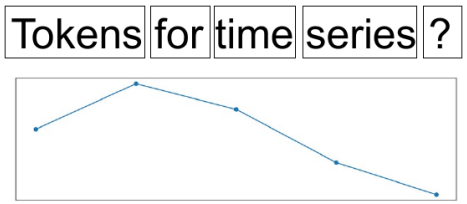


Dimensions: (tokens, channels)
Transformers architecture (Vaswani et al. 2017)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$$



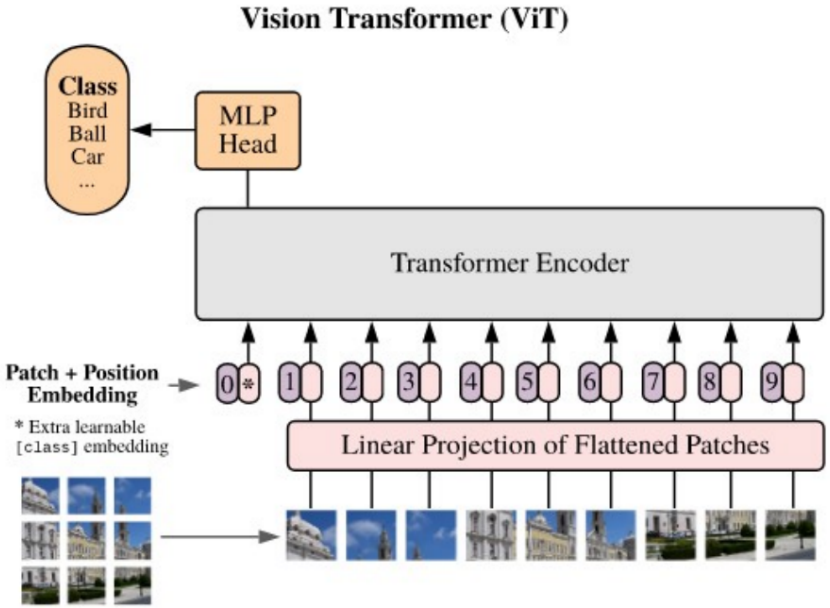
Self attention:
$Q = V = K$



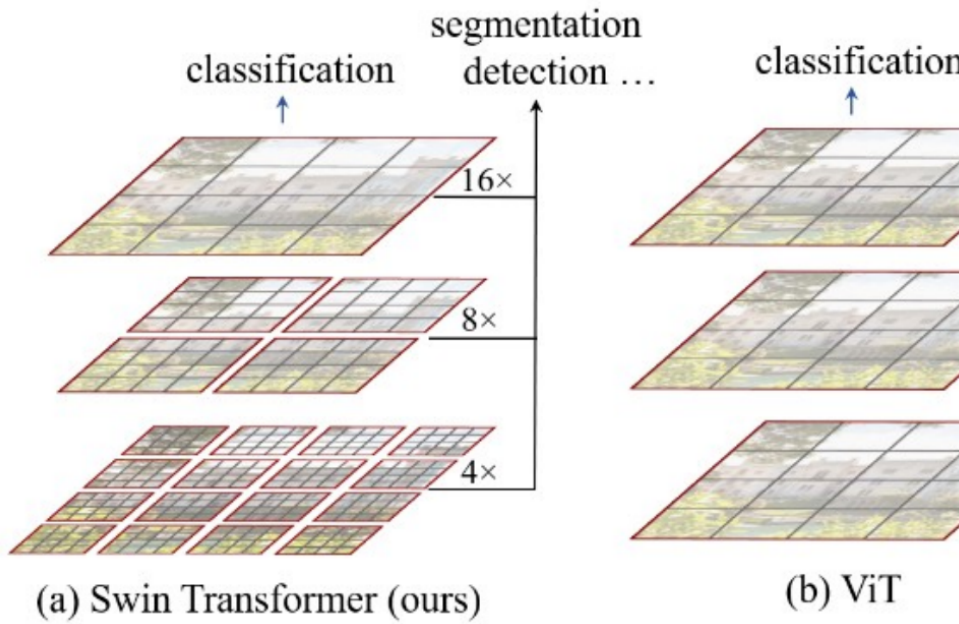
NLP tokens and an 1-D time series.

Transformers for images: ViT and Swin

From tokens describing words to tokens describing parts of an image.
These can be used with SimVP for spatiotemporal prediction!

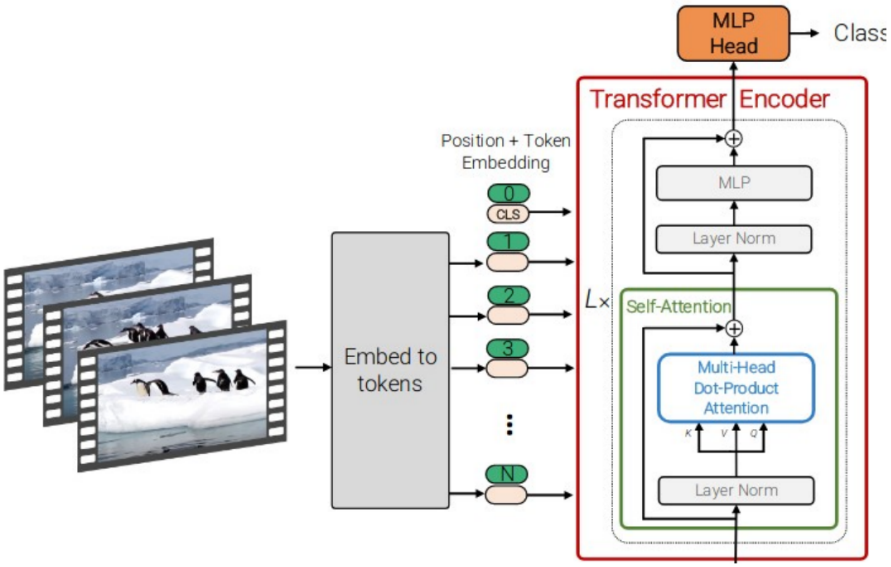


Vision Transformer (ViT): Attention is used to classify the entire image (DOSOVITSKIY et al. 2021)



Swin Transformer: Attention is used to describe the neighborhood (LIU et al. 2021)

Transformers for video: ViVit



Video Vision Transformer (ARNAB et al., 2021)

Similar to Vision Transformer (ViT), ViVit was designed to classify an entire video.

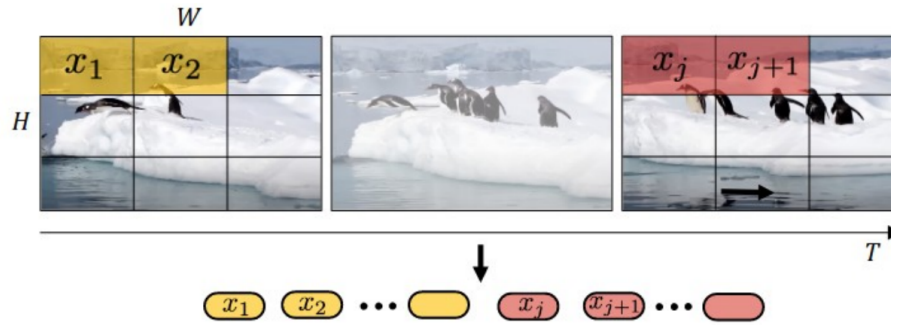


Figure 2: Uniform frame sampling: We simply sample n_t frame and embed each 2D frame independently following ViT [18].

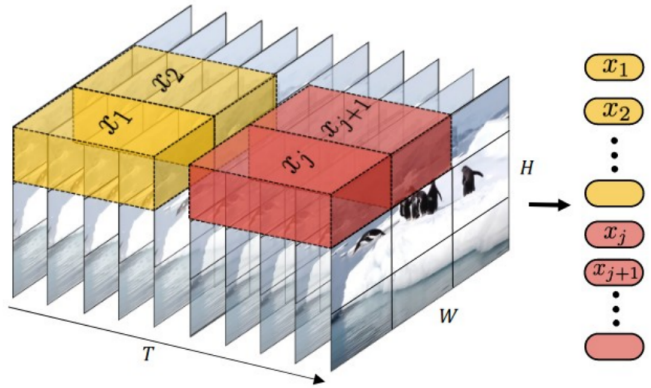
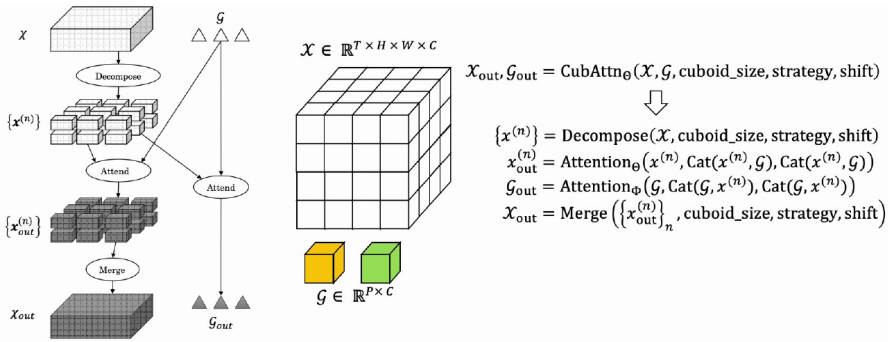


Figure 3: Tubelet embedding. We extract and linearly embed non-overlapping tubelets that span the spatio-temporal input volume. (ARNAB et al., 2021)

Transformers for spatiotemporal prediction

- Earthformer
  - Tokens: Cuboid attention
  - Global features



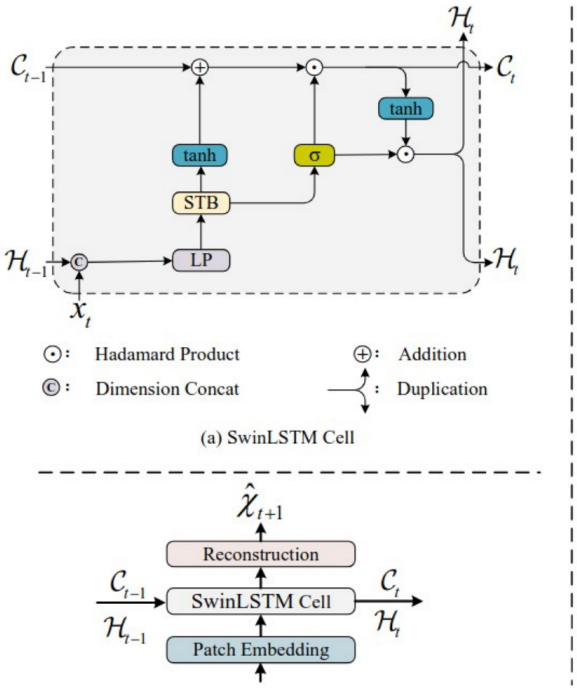
- MagVIT (MAsked Generative VIdeo Transformer)
  - Tokens: 3D-Vector-Quantized (VQ) auto-encoder
  - Masked token modeling
  - Multi-task learning

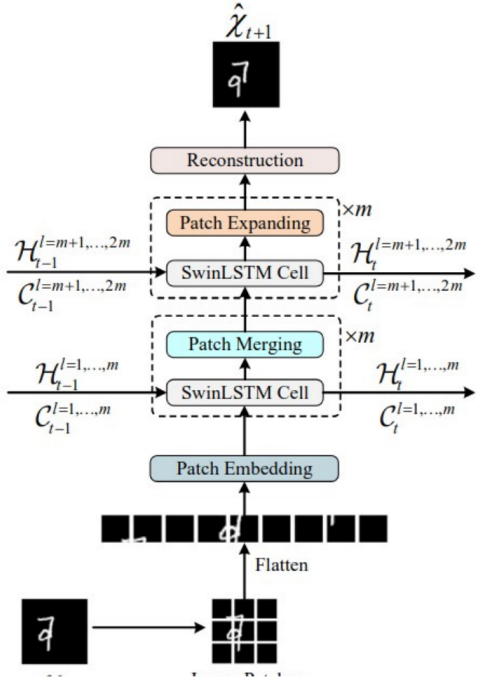
Earthformer (PyTorch):

`https://github.com/amazon-science/earth-forecasting-transformer` (Gao et al. 2022)

MagVIT (Google Jax): `https://github.com/google-research/magvit` (Yu et al. 2023)

Recurrent neural network (again?): SwinLSTM





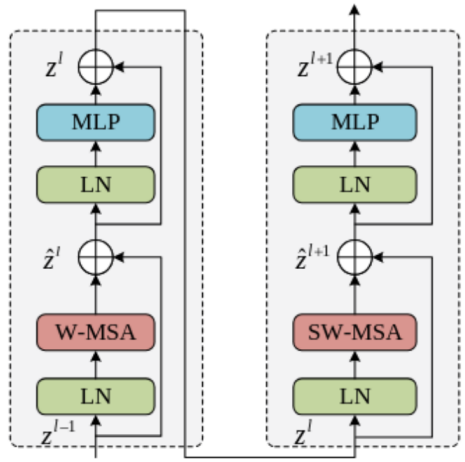


Figure 3. Swin Transformer Block.

Source: `https://github.com/SongTang-x/SwinLSTM`

Being ported to `https://github.com/chengtan9907/OpenSTL` (Tang et al. 2023)

Future? Structured state space sequence models (SSM)



- Linear or near-linear scaling in sequence length (transformer grows quadratically)
- Remove the nonlinearity in RNNs (remember LSTM gates?)
- Mamba: Linear-Time Sequence Modeling with Selective State Spaces (Gu et al. 2024) `https://github.com/state-spaces/mamba`
  - Selection Mechanism
    - Select data in an input-dependent manner
    - SSM parameters based on the input

Linear recurrence computation

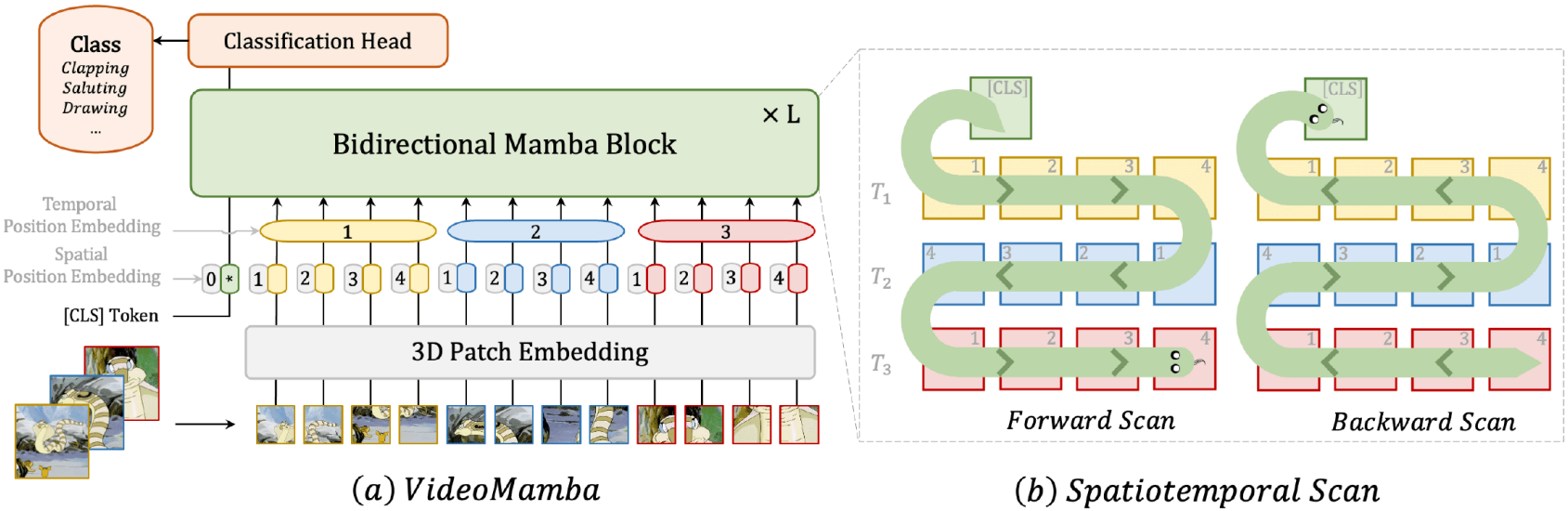
$$h_t = \overline{A}h_{t-1} + \overline{B}x_t$$
$$y_t = Ch_t$$

Global convolution computation

$$\overline{K} = (C\overline{B}, C\overline{AB}, \ldots, C\overline{A}^k\overline{B}, \ldots)$$
$$y = x * \overline{K}$$

Future? Structured state space sequence models (SSM)

- Video Mamba (Li et al. 2024):
  - `https://github.com/OpenGVLab/VideoMamba`
  - Designed for video classification, so we should have a forecast architecture soon.



(a) *VideoMamba* (b) *Spatiotemporal Scan*

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What changed since 2024?



- Global weather models became the main driver of the field
  - Graph networks and transformers on spherical meshes replaced ConvLSTM/UNet-style designs.
- From deterministic to **probabilistic** forecasts
  - Diffusion models and CRPS-trained ensembles instead of plain MAE/MSE training.
- **Foundation models**: pretrain on huge datasets, fine-tune for your task.
- AI models are now **operational** at weather services (ECMWF AIFS, Google WeatherNext).
- SSM/Mamba: competitive and efficient, but did not replace transformers.

Global weather forecasting models

Model	Architecture	Notes
GraphCast (Lam et al. 2023)	GNN: encoder → processor on an icosahedral multi-mesh → decoder	Beat ECMWF HRES on most targets
Pangu-Weather (Bi et al. 2023)	3D Earth-specific Swin transformer	Swin ideas applied to the atmosphere
GenCast (Price et al. 2025)	Diffusion model on a sphere mesh	Probabilistic ensemble, beat ECMWF ENS
Aurora (Bodnar et al. 2025)	3D Swin + Perceiver, foundation model	Fine-tuned for air quality, waves, cyclones
NeuralGCM (Kochkov et al. 2024)	Differentiable physics solver + learned components	Stable long (climate) rollouts
ECMWF AIFS / AIFS-ENS	GNN encoder/decoder + transformer processor, CRPS training	Operational ensemble since 2025
WeatherNext 3 (Google 2026)	Functional Generative Network (FGN) mesh transformer	Hourly runs, satellite and station data, 64 members

Caveat: physics-based IFS ensembles still beat raw AIFS for 10-m wind at stations in a 2026 comparison.

Generative and probabilistic forecasting

Why?

- MAE/MSE training produces **blurry** forecasts (the mean of possible futures).
- Decisions need uncertainty: ensembles, probabilities of extremes.

How?

- Diffusion / flow matching
- CRPS loss with noise injection (AIFS-ENS, FGN)
- Deterministic backbone + generative residual

Nowcasting (precipitation, radar)

- MetNet-3: transformer with dense supervision
- DGMR, NowcastNet: GAN-based, physics-conditioned
- DiffCast, CasCast, PreDiff: diffusion keeps small-scale details

New evaluation metrics

- CRPS, spread–skill ratio
- Power spectra (does it blur?)

Generic spatiotemporal models and station data

Video-like prediction (OpenSTL lineage)

- SimVP / TAU / gSTA: still strong, efficient baselines
- PredFormer (2024): pure transformer with factorized space-time attention, beats recurrent models
- Mamba variants (VMRNN, ST-Mamba): efficient on long sequences

Stations and graphs

- Spatiotemporal GNNs / transformers for sensor networks (traffic, air quality)
- Time series foundation models for zero-shot forecasting per station: TimesFM, Chronos, Moirai, TimeGPT

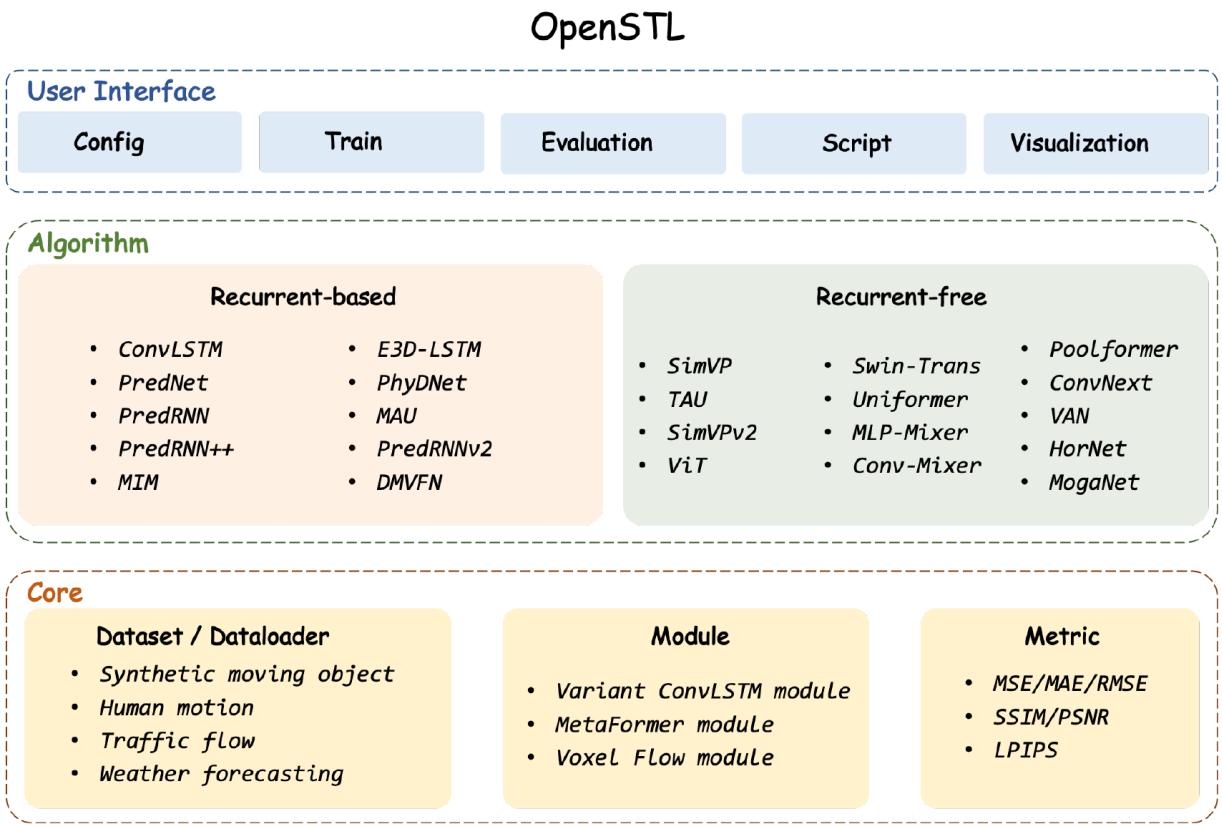
Takeaway for global maps (e.g. GIMs)

Spherical meshes, probabilistic training and pretrained models are worth testing before designing a new architecture.

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OpenSTL (Pytorch+lightning)



Possibilities

- Create your dataloader and train a model.
- Use the models in your project.
- Create new models.

What's the difference?

- Learning capacity
- Memory
- Forecast procedure
- Training speed
- Prediction speed

(Tan et al. 2023)

OpenSTL benchmark datasets (results)

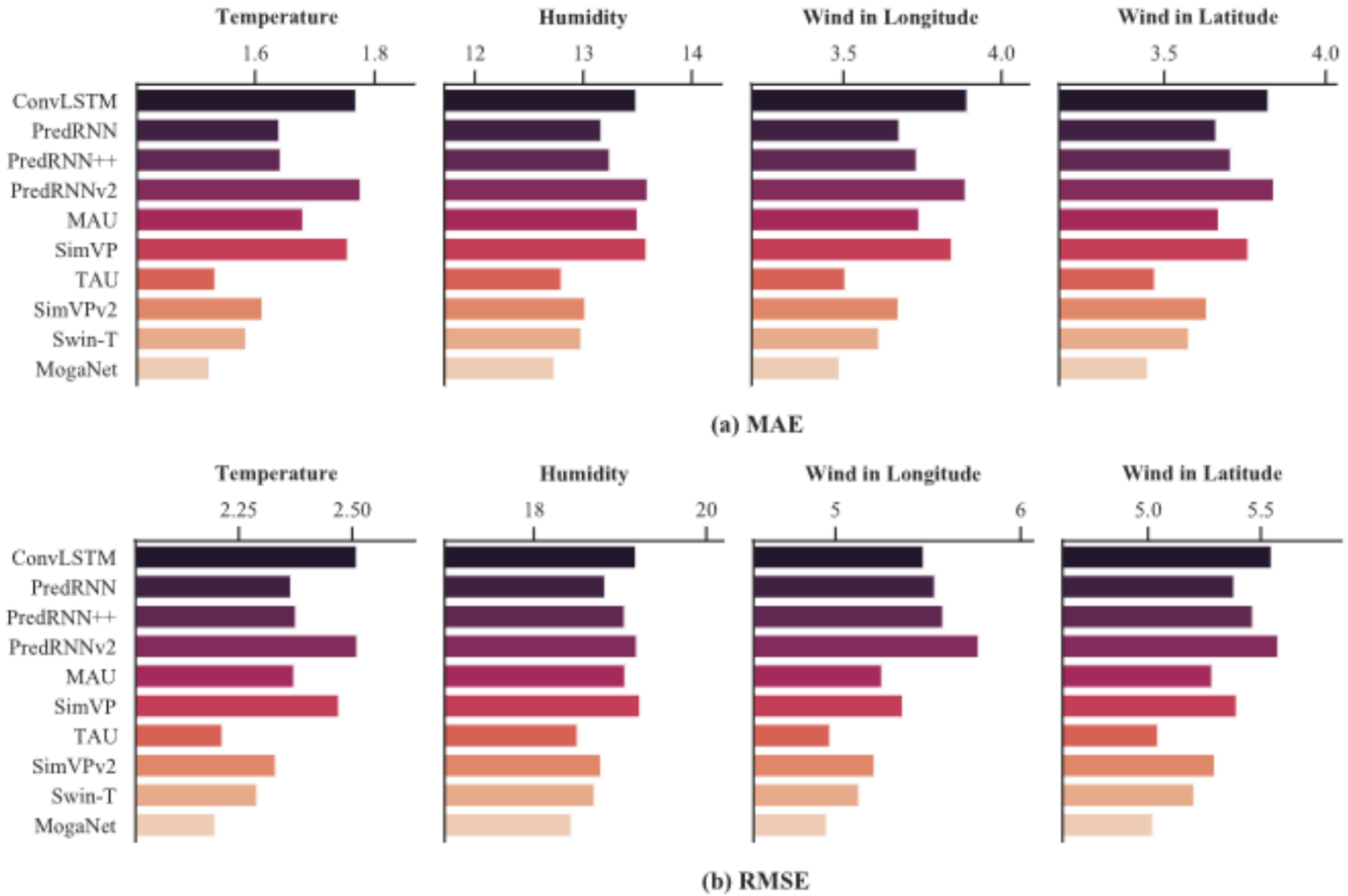


Figure 17: The (a) MAE and (b) RMSE metrics of the representative approaches on the four weather forecasting tasks in WeatherBench (Muti-variable setting).

(Tan et al., 2023)

Google Colab code



- **OpenSTL in Global Ionosphere Maps prediction (hands on)**
  - https://github.com/mauriciodev/tec_forecast/blob/main/examples/OpenSTL_GIM.ipynb
- OpenSTL custom dataloader tutorial
  - https://github.com/chengtan9907/OpenSTL
  - https://colab.research.google.com/drive/19uShc-1uCcySrjrRP3peXf2RUNVzCjHh?usp=sharing
- GIM Forecast with PyTorch (Paulo et al, 2023)
  - https://github.com/mauriciodev/tec_forecast/blob/main/extra/tec_forecast_torch.ipynb
- GIM Forecast with Tensorflow/Keras (Paulo et al, 2023)
  - https://github.com/mauriciodev/tec_forecast/blob/main/examples/tec_forecast_tf2_lab.ipynb

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Loss functions and evaluation metrics



Numerical prediction (Regression)

$$MAE = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

$$Huber = \frac{\sum_{i=1}^{n} L_{huber}(y, \hat{y})}{n}$$

where

$$L_{huber}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| < \delta \\ \delta\left(|y - \hat{y}| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases}$$

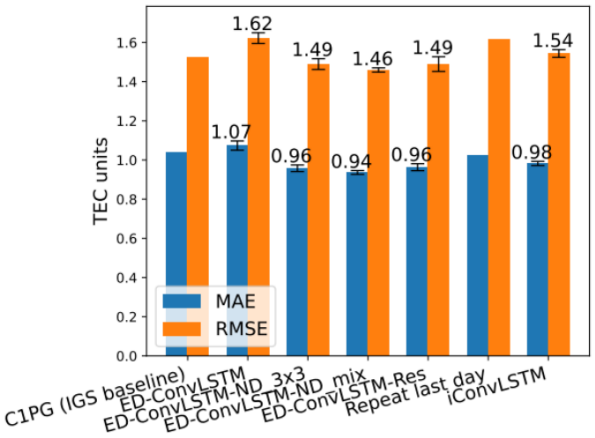
$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

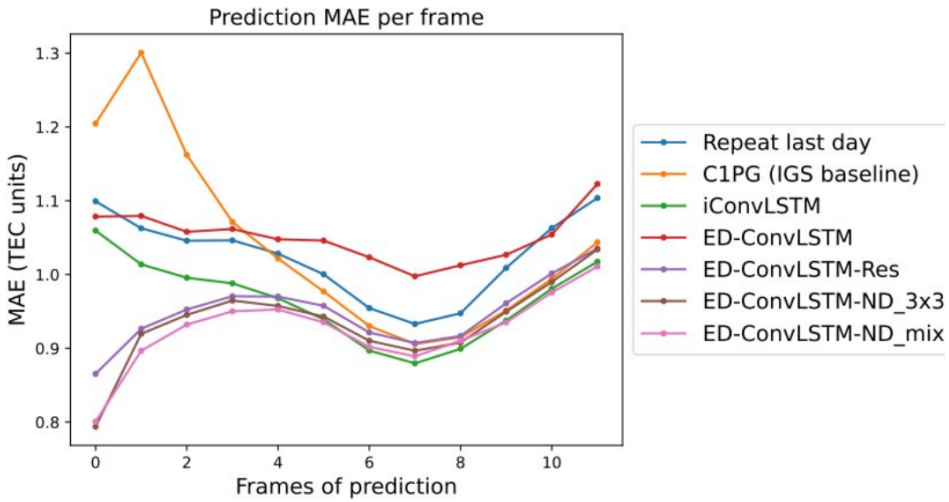
Categorical prediction

- Categorical cross entropy loss
- Focal loss
- Precision, recall, F-score
- Intersection over union (IOU)

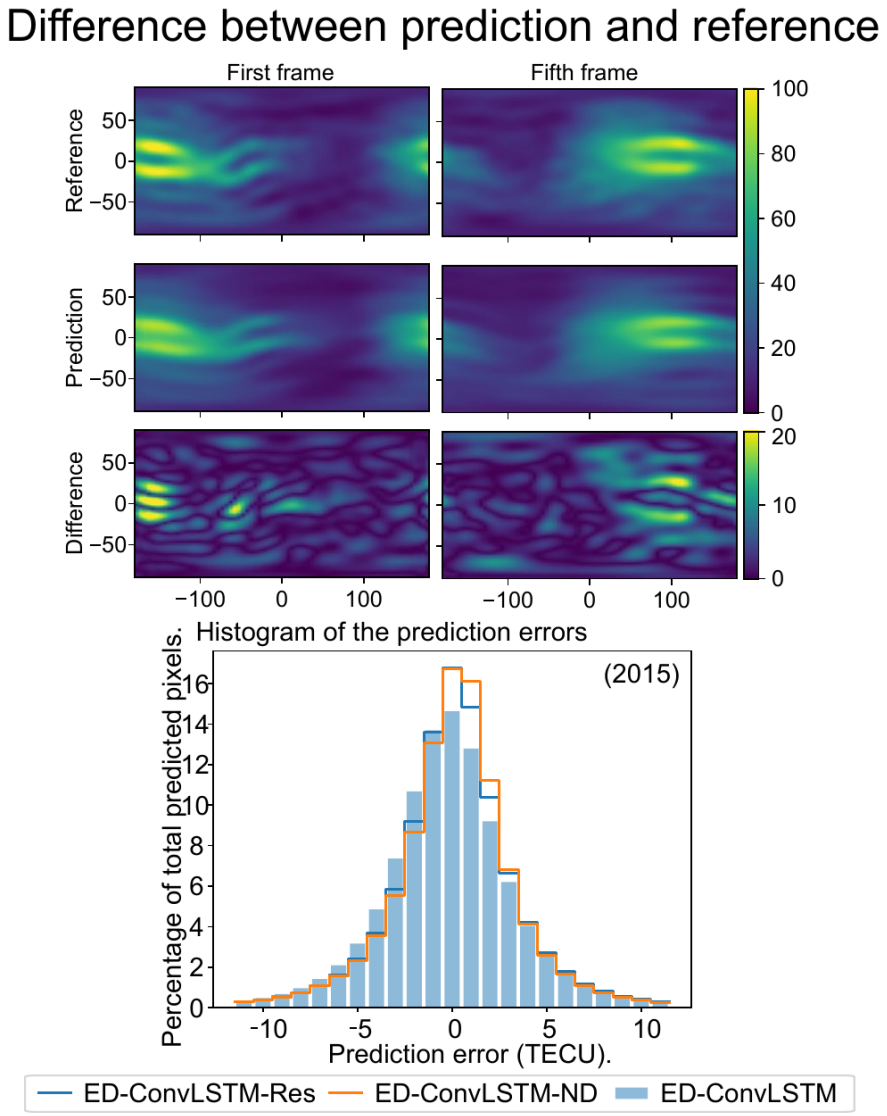
n: Number of samples in a batch
y: Values from the training sample;
$\hat{y}$: Values predicted with the network.

Exploring the results



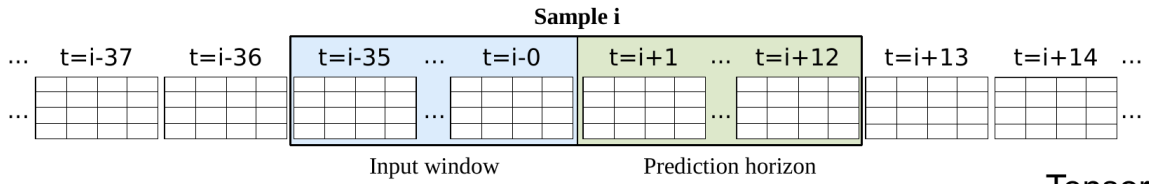


Frame-wise MAE.



Data set construction: What do we measure?

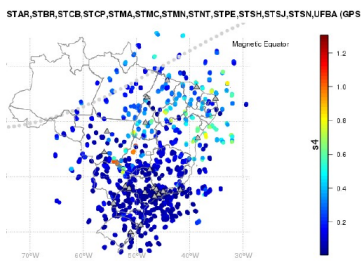
This is what we want:



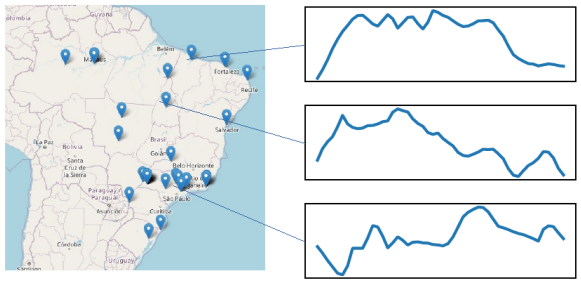
Tensor dimensions: (t,c,x,y)

Pandas: Temporal resampling
GDAL or PyKrige: Spatial interpolation

But sometimes this is what we have:



Scattered points with a timestamp.

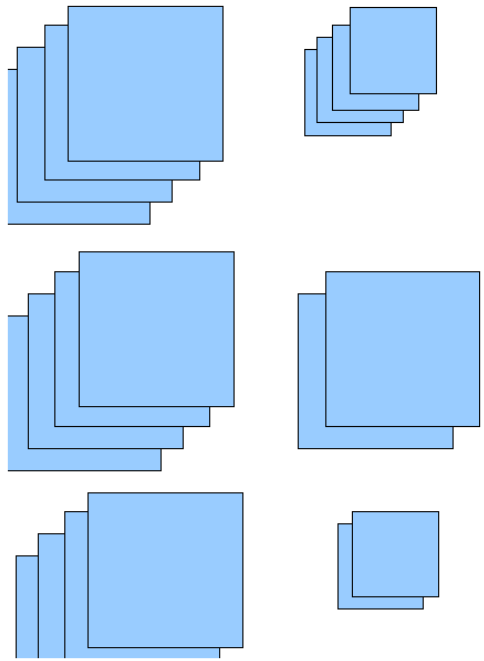


Time series on multiple stations.

Among others (1D? Graphs?)

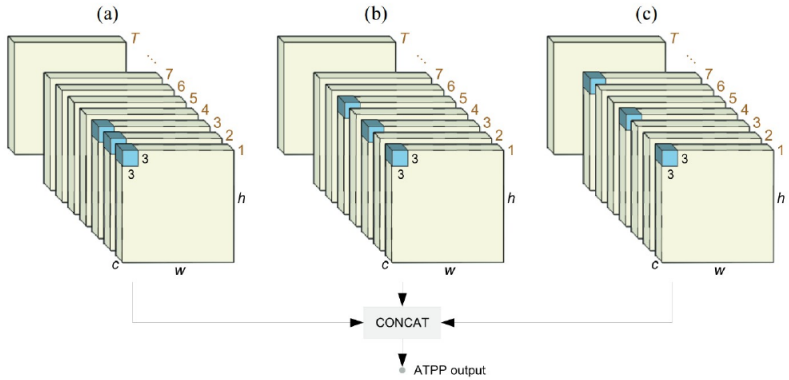
Convolutional layers dimensions

- Popular pooling options:
  - Max
  - Average
- Spatial pooling
  - $(t, x, y, c) \rightarrow (t, x/2, y/2, c)$
- Temporal pooling
  - $(t, x, y, c) \rightarrow (t/2, x, y, c)$
- Spatiotemporal pooling (uncommon)
  - $(t, x, y, c) \rightarrow (t/2, x/2, y/2, c)$
- **Convolution with stride can change resolution too.**
- Spatial and temporal resolution changes must make sense for the event.

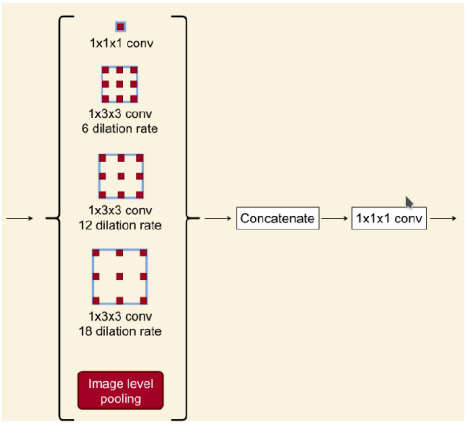


Convolutional layers dimensions

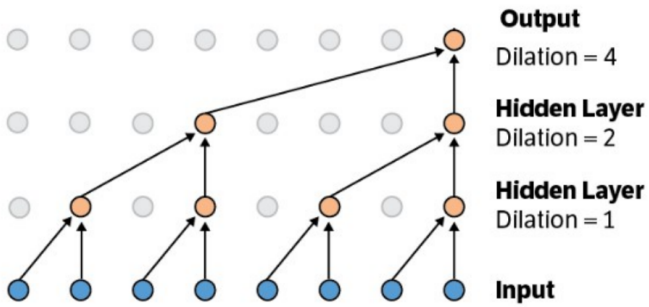
- **Dilated convolution**
  - Spatial
  - Temporal
  - Spatiotemporal (uncommon)
- **Deformable**



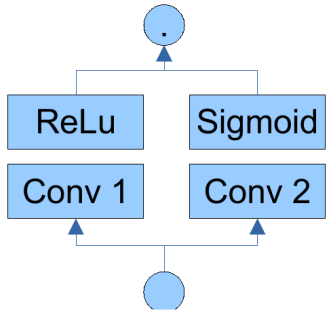
UNet 3D Atrous temporal pyramid pooling (Rogozinski, 2021)



Broad-UNet Atrous spatial pyramid pooling (Fernández, 2021)



Wavenet's sequence of time dilated convolutions (Borovykh, 2018)



Sigmoid gates control the output of a block.

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